# Distributed Execution for K3-Class Models

Deep Dive into Kimi K3 — Chapter 38

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Goal: name every rank, group, layout, and owner; refine TP, PP, CP, and EP operator by operator; close class gradients under one objective; and keep cost and K3 claims in their evidence scope.

# Roadmap

Typed axes, groups, and owners

Executing one step

Keys, resources, and oracles

# Typed axes, groups, and owners

# One row, five typed crossings




- 32-rank fixture $r = 16d + 8p + 4c + 2e + t$; rank 23 sits in TP $(22, 23)$, EP $(21, 23)$, CP $(19, 23)$, PP $(23, 31)$, DP $(7, 23)$. [DERIVED: chapter construction]

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- A group is identity, ordered members, and epoch; a layout has six fields; owner, holder, executor, destination, and update owner are distinct roles.

# Executing one step

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- **Closure:** sums only until owner 5 divides 384 by $Z = 64$: $g = 6$, $\theta = 10 \to 9.4$, revision published.

[DERIVED: exact chapter witnesses] [REPORTED: K3 Sec. 5.2: PP/VP, EP, CP, ZeRO-1, Pipeline ZeRO-2]

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Diagnose in dependency order. Next: Chapter 39, Memory-Efficient Training, takes the event graph and its final-consumer lifetimes.